
Hidden-Feature Matching and Left-Right Movement: Different Outcomes in a Gait JEPA

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Abstract

1 We test whether learning to predict hidden skeleton features helps estimate a
2 left-right difference in movement speed during walking. A small skeleton Joint-
3 Embedding Predictive Architecture (JEPA) learns without movement labels; a
4 separate regression then estimates a signed speed contrast from its frozen fea-
5 tures. In source-video cross-validation on 625 clips from 93 videos, trained pre-
6 dictors more consistently favor hidden features from the correct clip over features
7 from another video. With the same expanded feature summary and regression
8 procedure, however, mean source-balanced R^2 falls from 0.22 at initialization to
9 0.10–0.11 after training. Motion-weighted and connected-region masks leave their
10 movement-readout benefits uncertain. The comparison identifies a gap between
11 hidden-feature matching and access to a specified observable of an evolving gait
12 system, within this cohort, training recipe and readout. It motivates evaluating
13 observable movement alongside latent prediction in representations intended for
14 temporal modeling.

15 1 Introduction

16 Walking involves coupled body landmarks whose positions change over time. Their trajectories carry
17 information about movement that a single pose cannot supply, while video-derived coordinates cap-
18 ture only part of the physical process: camera geometry and pose errors affect them, and forces and
19 contact remain unobserved. Gait rhythm, phase and coordination make this an appealing temporal
20 system to study. Temporal world modeling aims to represent an evolving system well enough to pre-
21 dict its observable behavior; our experiment examines access to one movement measurement within
22 that broader objective.

23 We use a small S-JEPA [1] to predict features at hidden positions in a skeleton sequence, then ask a
24 separate regression to estimate a left-right contrast of median coordinate speeds. The central null is
25 no improvement from JEPA training over matched initial weights on held-out source videos. This
26 null also motivates the registered reflection test; the mask comparisons and expanded readout were
27 developed on the same cohort and remain exploratory. By comparing the training task with a mea-
28 surement of the observed movement, the study offers focused evidence for evaluating representations
29 used in temporal foundation models.

30 2 Data, prediction task and evaluation

31 **Measuring movement.** Quality checks retain 625 GAVD clips from 93 source videos [2]. Archives
32 contain $T_i \times 33 \times 4$ arrays of three estimated coordinates and visibility, plus frame numbers and
33 frame rate. The coordinates are image normalized with inferred depth, rather than calibrated metric
34 positions. We center observed coordinates at the pelvis and divide by body width, keeping original

timestamps and leaving gaps unfilled. Five pairs—shoulders, knees, ankles, heels and foot tips—each need at least eight transitions with both sides observed at both ends. For left and right median coordinate speeds $m_{L,k}, m_{R,k}$, the signed contrast is

$$y = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}.$$

Reflection M flips the centered horizontal coordinate and exchanges anatomical sides, including validity flags, so $y(Mx) = -y(x)$. Hips establish the pelvis but are excluded from the sum because centering equalizes their speed magnitudes. The contrast measures relative median speed. It omits phase, stride timing and loading, and remains unchanged when a trajectory and its sampling intervals are reversed. Its recovery alone therefore provides limited evidence about temporal dynamics.

Preparing encoder inputs. A landmark is observed when visibility is at least 0.45 and its coordinates are finite. A separate path fills interior gaps of up to four samples and treats those positions as valid. After pelvis centering and body-width normalization, coordinates and validity are linearly resized by sequence index to 64 positions; original time intervals are discarded. Resized validity must reach 0.999, and invalid coordinates are zero. Each joint’s four-position patch projects 12 coordinates to 96 features, producing 16×33 tokens. All four positions must be valid for a patch to be used. Anatomical identities and prepared positions remain, but original duration and cadence are not retained; trajectory-derived quantities can also change.

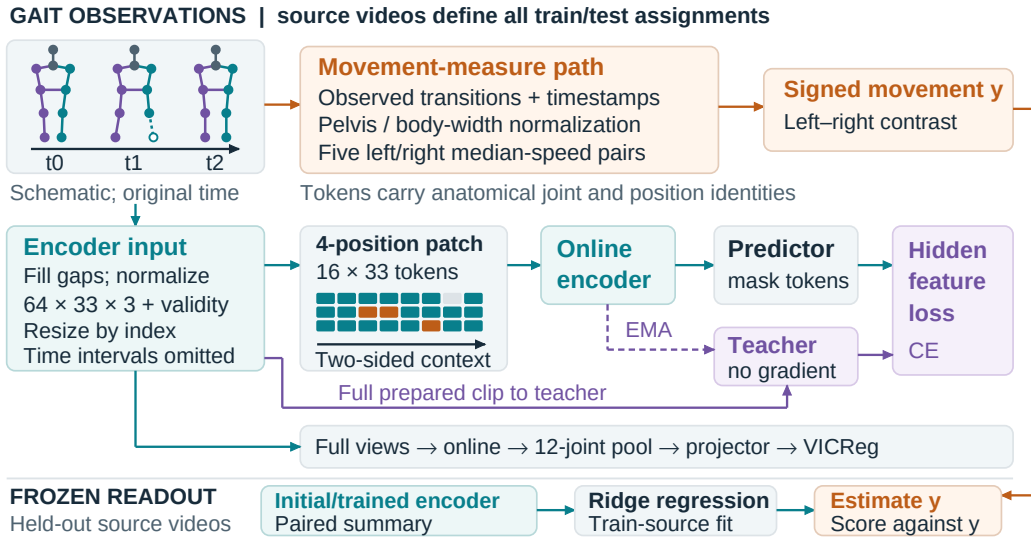


Figure 1: Completed model. Poses and tokens are schematic: skeleton colors mark anatomical left (teal) and right (purple), with an open circle for a missing landmark. Tokens are visible (teal), hidden (orange) or missing (gray). Scattered/motion-weighted and connected-region masks may leave context on both sides. Full clips reach the teacher and regularizer; target y supervises the separate readout.

Learning hidden features. The online encoder zeroes hidden patch content before adding joint and position embeddings. The predictor inserts learned mask tokens at target positions and estimates the teacher’s features there. Cross-entropy compares distributions over feature channels at hidden valid positions, averaging within clips and then across clips. The teacher receives the full prepared clip; its weights follow an exponential moving average of the online weights without gradients. A VICReg regularizer [3], weighted 0.05, also uses full clips: two geometric views pass through the online encoder, a twelve-joint gait pool including hips, and a projector. Movement targets and gait-category labels never enter these losses.

The grid contains 125 encoders: three motion-mask conditions and two region conditions across five folds and five seeds, each trained for 1,200 AdamW updates in CUDA BF16 and evaluated in FP32. Motion conditions use random, MAMP-style [4] or mixed uniform/clipped-motion sampling over all

33 joints, with counts derived from a twelve-joint gait pool. Region masks hide six connected landmarks over eight of sixteen blocks. Each family has its own count-matched random reference and shares initial weights, source draws, views and update counts. Reflection augmentation and explicit reflection loss are absent from this grid. Appendix A gives settings and their separate evidence.

Estimating movement from frozen features. Five outer folds keep every video’s clips, overlapping windows, mirrors and augmented views together wherever these derivatives are used. A fresh encoder fits only outer-training sources, sampling videos equally. Its matched initialization and final encoders are then frozen for ridge regression, a linear readout with a coefficient penalty. Three inner source folds select that penalty using outer-training labels only. Imputation and scaling fit within each inner training split, then refit with the selected regression on all outer-training sources. Inner-validation clips may have contributed without labels to encoder fitting.

The readout summarizes five bilateral pairs using feature means (960 inputs), expanded with standard deviations, absolute adjacent-block changes and ten support fractions (2,890 inputs). We pool held-out predictions within each seed, give each source video equal total weight, and average the five R^2 scores. The 2,000 paired source-bootstrap resamples retain each video’s clips and seed predictions together. Their 95% intervals condition on fitted models and splits, excluding retraining, partition and development-selection uncertainty. Repeated seeds and mask draws reuse the same 93 videos.

3 Feature matching and movement estimates give different results

With the expanded summary, initial features yield mean $R^2 = 0.223$, compared with 0.101–0.114 for trained teachers (Figure 2). Every training condition scores below its initialization in every seed; all five paired intervals are below zero (Table 1). Mean absolute error increases by 0.0020–0.0025 in the dimensionless contrast, from 0.0415 to 0.0436–0.0440. Final online encoders also score below their initial weights.

Training mask	Mean R^2	ΔR^2	95% source interval
Random (motion)	0.114	-0.109	[-0.170, -0.041]
MAMP-style	0.113	-0.110	[-0.165, -0.051]
Motion mixture	0.114	-0.108	[-0.171, -0.042]
Random (region)	0.109	-0.113	[-0.166, -0.060]
Connected region	0.101	-0.122	[-0.183, -0.055]

Table 1: trained teacher minus matched initialization, using the expanded summary. Exploratory intervals resample source videos with fitted models fixed; differences precede rounding.

Expanding the initial summary raises R^2 from 0.071 to 0.223 (+0.152). This combines several changes: more regression inputs, temporal variation, absolute feature increments and observation support. Standard deviations discard order, and absolute increments discard direction, so the gain cannot be attributed specifically to temporal order. The initial control already includes a pretrained pose detector, preparation, anatomical identities and supervised regression.

The predictor diagnostic compares source-weighted mean squared feature error against correct-clip targets and valid targets from another source, holding clips and hidden positions fixed. Correct targets have lower error in 375/375 trained checks and 33/75 initial checks. These totals cover 125 trained models and 25 deduplicated initial controls, each under left-leg, right-leg and scattered gaps. Pose, viewpoint and missingness shared by visible context and full-clip teacher features could support this correspondence alongside motion. Changing teachers and target distributions also prevent raw training losses from providing a common ranking of representation quality.

Mask construction works in the inspection draws: motion weighting selects higher-motion targets, and regions reduce targets with both immediate temporal neighbors visible from 69.9% to zero. Motion masks hide approximately 17% of valid tokens and regions 9.9%, requiring distinct random references. Their movement-readout intervals cross zero, leaving mask benefits uncertain without establishing equivalence (Appendix A).

Input preparation also changes the observable. On 623 clips from 92 sources with finite values on both paths, a retained calculation comparison gives 70.4% source-weighted sign agreement and

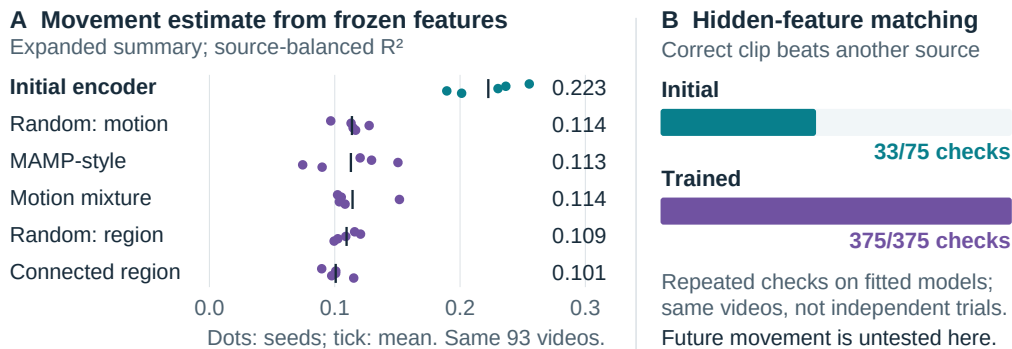


Figure 2: Different outcomes from the same fitted models. A: movement estimates from the matched initial encoder and trained teachers with the expanded summary; dots are seed scores and ticks their mean. B: correct-clip feature error is lower in 375/375 trained checks and 33/75 distinct initial checks. These checks reuse models and videos.

108 $R^2 = 0.218$. This disagreement between measurement paths neither bounds achievable model
109 accuracy nor isolates the effect of an individual preparation step.

110 4 Implications and a future-motion test

111 For this gait measurement, greater consistency in hidden-feature matching accompanies poorer
112 movement estimates under the expanded readout. Another readout may access information that ridge
113 regression leaves unused: 49/125 teacher fits and 96/125 online fits select the largest tested penalty,
114 10,000. Wider training-only penalty selection, support-only controls and summaries of matched
115 dimension would help distinguish readout limitations from changes in the features. Source-video
116 holdout leaves participant independence, chronological generalization and an untouched develop-
117 ment cohort unresolved; recording and pose-estimation cues also remain possible explanations.

118 The next experiment would predict future coordinates from an observed prefix, with preparation,
119 normalization and mask selection restricted to the past. Training evidence for past-feature read-
120 out and predicted-future-feature decoding is currently synthetic; real-data horizon counts establish
121 eligibility only. A real-data comparison should use common valid endpoints and physical-time
122 horizons, with last-position, constant-velocity, direct-pose, initial-encoder and mismatched-training-
123 future controls. Decoders and their settings must use training sources only, and future-coordinate or
124 visibility changes must leave the prefix and forecast unchanged. Observed future-teacher features
125 provide a decoder diagnostic; only features predicted from the past test forecasting. Several horizons
126 from one prefix still require a separate recursive-rollout evaluation.

127 The present result concerns one small gait model, cohort and readout. Foundation-model scale,
128 broad transfer, persistent temporal state and a working simulator require evidence beyond this study.

129 5 Related work

130 Prediction targets and available context influence which movement differences a representation re-
131 tains. Skeleton methods predict learned features [1] or explicit motion [4], while connected-body
132 masks reduce nearby cues [5]. Their action-recognition evaluations assess suitability for labels.
133 Video anticipation and action-conditioned planning [6] ask different questions about behavior, and
134 separating invariant from transformation-sensitive features [7] makes this dependence on the in-
135 tended task explicit. Our paired initial-versus-trained comparison examines a particular observed
136 gait quantity within that research agenda. The mask variants borrow sampling ideas from related
137 methods, whose complete systems are not experimentally compared here.

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A Implementation and supporting checks

Of 666 selected annotations, 642 match pose archives and 625 pass quality checks. Archive coordinates are $p_x = (x_0 + uW_c)/W$, $p_y = (y_0 + vH_c)/H$ and $p_z = dW_c/W$, for crop origin (x_0, y_0) , crop size W_c, H_c , frame size W, H and pose estimates u, v, d . Body scale is the median valid shoulder/hip width, with unit fallback. The target requires observed hips; the encoder substitutes the clip’s median valid pelvis midpoint when hips are missing (zero if none exists). Gap filling is linear between observed endpoints, without boundary extrapolation.

Training uses width 96, encoder/predictor depths 4/2, four heads, batch 20, AdamW learning rate 0.001, weight decay 0.05, gradient clipping at 1 and teacher EMA 0.999. Geometric views rotate up to 8 degrees in the horizontal-depth plane and translate up to 0.03 in the image plane; the teacher receives unaugmented input. Teacher/predictor temperatures are 0.06/0.10, with center EMA 0.9. The regularizer’s two-layer 96-dimensional projector uses GELU; internal invariance/variance/covariance weights are 25/25/1. The motion target count is half the batch’s smallest valid twelve-joint gait pool, rounded down, sampled across 33 joints. Motion weights use prepared-index displacements: MAMP-style temperature is 0.8; the mixture combines 25% uniform with 75% motion weights capped at the 95th positive-score percentile.

Summary means and standard deviations use common paired validity; increments require consecutive common valid blocks. Missing support produces zero values. Each pair contributes six 96-vectors (left/right differences and sums) and two support fractions, totaling 2,890. Ridge candidates are 0.01, 0.1, 1, 10, 100, 1000, 10000. For V videos with n_v clips, $w_{vi} = 1/(Vn_v)$ and $\bar{y}_w = \sum w_{vi}y_{vi}$ define

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi}(y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi}(y_{vi} - \bar{y}_w)^2}.$$

The test mean is a metric reference; a deployable constant prediction must be fitted on training sources. Test folds contain 19/19/19/18/18 videos and 189/182/72/77/105 clips. Retained audits check archive integrity and split adherence. Missing person IDs and acquisition records leave cross-upload/person deduplication and exact cohort reconstruction unresolved.

MAMP-style, motion-mixture and region masks minus their own random references give $\Delta R^2 = -0.0008, +0.0005$ and -0.009 , with intervals $[-0.020, 0.015]$, $[-0.015, 0.015]$ and $[-0.033, 0.018]$. A limited 264-value coordinate/validity baseline reaches $R^2 = 0.035$. Grid intervals are exploratory percentiles (bootstrap seed 812), without familywise or development-selection adjustment. The supplement retains aggregate seed scores and recorded intervals. Raw predictions and checkpoints were unavailable for this revision, which checks aggregate consistency rather than rerunning inference or source bootstraps.

Reflection augmentation has separate real-data summary evidence (mirror probability 0.5, scheduled EMA, compact readout, four inner folds): relative to training without mirrors, normalized teacher-token reflection error decreases by $\Delta q = -0.008 [-0.010, -0.007]$, while $\Delta R^2 = 0.004 [-0.006, 0.013]$ leaves the movement benefit uncertain. The comparison swaps joint identities while leaving feature channels unchanged. An explicit reflection loss has synthetic evidence only and adds two full encoder passes. The construction $g(x) = [h(x) - h(Mx)]/2$ guarantees sign reversal independently of accuracy.

Clinical asymmetry motivates this observable [8], but pair contrasts can cancel and affected-side labels and clinical validation are absent. Videos and derived poses are excluded from the supplement; public annotation access does not authorize their redistribution.

AI assistance. An AI assistant supported author-side evidence reconciliation, literature checking, revision, figure programming and adversarial checks. No new training was performed. Authors remain responsible for the text, figures, claims and references.